



OPEN An Arabic language benchmark and hybrid BERT–BiGRU–CapsNet model for optimism and pessimism detection

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Accurate identification of optimism and pessimism has important applications for individual health and societal well-being. Despite the growing interest in sentiment analysis and emotion detection, limited research has focused on the detection of optimism and pessimism, particularly in the Arabic language. Existing studies primarily address general sentiment classification, leaving a significant gap in modeling optimism and pessimism detection. To the best of our knowledge, this study presents the first effort to introduce a dedicated Arabic dataset for optimism and pessimism analysis, comprehensively evaluate the dataset on seven Arabic language models, and propose a novel hybrid approach for detecting optimism and pessimism. In this paper, we introduce the Saudi optimism and pessimism detection (SOPD), the first and largest dataset for detecting optimism and pessimism in Arabic. It was constructed from posts and replies collected from the Hawamr and Kooora forums, focusing on Arabic-language content in the domains of finance and sports. The dataset comprises a total of 16,152 samples, manually annotated into three sentiment classes: optimistic, pessimistic, and neutral. The dataset was then systematically evaluated using seven Arabic language models: Arabic-BERT, CamelBERT-DA, ARBERTv2, AraBERTv02, QARiB, MARBERTv2, and SaudiBERT. Additionally, we proposed a novel hybrid deep learning architecture, BERT-BiGRU-CapsNet, which combines a Bidirectional GRU (BiGRU) layer with a Capsule Network (CapsNet) on top of a BERT encoder. The evaluation results showed that SaudiBERT achieved the highest F1-Score of 93.11%, followed by MARBERTv2 with the second-highest F1-Score of 87.25%. Nonetheless, the proposed hybrid model, BERT-BiGRU-CapsNet, outperformed all traditional Arabic models with an F1-Score of 93.34%. The results provide valuable insights into the effectiveness of current Arabic models in classifying optimism and pessimism, offering practical directions for improving context-aware sentiment analysis in Arabic NLP. Moreover, this study holds significant implications for enhancing individual well-being, predicting behavioral trends, and informing the design of targeted mental health and social interventions. It also might enable businesses and enterprises to effectively analyze customer behavior, supporting strategic decision-making, product development, and customer engagement.

Keywords Optimism and pessimism, BERT-BiGRU-CapsNet, LLM, Arabic language models

With the rise of social media platforms, individuals increasingly express their thoughts, emotions, opinions, and experiences openly and frequently. Social media serves as a dynamic space for sharing optimism, pessimism, joy, frustration, and concern in real-time. The accessibility and openness of these platforms enable individuals to articulate their emotions, making it possible to identify key aspects of their personality. Daily written expressions often reflect underlying emotional states and can significantly influence interpersonal communication and behavioral interpretation. These expressions frequently convey psychological constructs that encompass emotional and behavioral dimensions such as pessimism and optimism. Identifying these constructs is critical for understanding and predicting human behavior, emotional responses, and decision-making across various domains. Indeed, accurately identifying optimism and pessimism plays a vital role in promoting individual well-being, predicting behavior, and analyzing social dynamics. Sentiment analysis (SA), a fundamental task in NLP, focuses on determining the polarity of opinions expressed in text as positive, negative, or neutral. While SA techniques are widely used across different applications, they often fall short in reliably detecting nuanced

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psychological states such as optimism and pessimism, particularly in linguistically complex or low-resource languages like Arabic.

Optimism and pessimism significantly influence both the psychological and physical health of individuals. Optimism is a personality trait that reflects the extent to which individuals hold generalized favorable expectations about their future¹. As an individual-difference variable, it indicates how strongly people expect positive outcomes in their lives. Higher levels of optimism are prospectively associated with greater subjective well-being, particularly during periods of adversity or challenge. Empirical evidence shows that optimism correlates with proactive health-protective behaviors, while pessimism is linked to behaviors that may harm health. Thus, optimism is consistently associated with indicators of better physical health¹. Pessimism, by contrast, is a personality trait marked by a consistent expectation of negative outcomes¹. It often correlates with negative emotional states such as depression and anxiety, whereas optimism tends to mitigate stress and promote improved physical health and overall well-being. These two modes of thinking are distinct, yet are best conceptualized as a continuum with varying degrees of emotional impact. Among individual-difference variables, optimism and pessimism stand out as strong predictors of happiness², stress regulation¹, and both mental³ and physical well-being⁴. Thus, modeling optimism and pessimism through social media activities presents valuable opportunities for enhancing individual health and promoting societal well-being, particularly in low-resource and linguistically complex languages such as Arabic. However, there remains a noticeable lack of datasets and research focused on modeling optimism and pessimism in Arabic, despite the growing presence of Arabic content on social media platforms.

Arabic ranks as the fourth most widely used language on the Internet, with over 400 million speakers⁵, and serves as the official language in 22 countries⁶. The language presents considerable complexity due to its rich morphology, diverse dialects, and the use of negation and intensifiers^{7,8}. It is written from right to left using a cursive script where most letters change their shape depending on their position within a word. Arabic is also highly agglutinative, meaning that prefixes, suffixes, and pronouns are often attached directly to the core word. This results in an enormous number of unique tokens, which diminishes the statistical weight of features and makes it harder for classifiers to generalize across different datasets. These linguistic features significantly affect the detection of optimism and pessimism. Consequently, Arabic's root-based morphology, along with its extensive system of negation, intensifiers, and modifiers, allows for highly nuanced sentiment expression that requires sophisticated modeling to decode. The coexistence of Modern Standard Arabic (MSA) with various regional dialects further complicates sentiment analysis. In addition, contextual cues, cultural nuances, and idiomatic expressions demand context-aware interpretation. As a result, advanced natural language processing techniques and specialized models are necessary to accurately identify and classify sentiments in Arabic.

To the best of our knowledge, this study is the first to introduce an optimism and pessimism dataset in Arabic, evaluate it across seven Arabic language models, and propose a hybrid deep learning architecture, BERT-BiGRU-CapsNet, which integrates a Bidirectional GRU (BiGRU) layer and a Capsule Network (CapsNet) on top of a BERT encoder for optimism and pessimism classification. The primary objective of this hybrid model is to enhance hierarchical feature modeling and inter-class representation learning. Specifically, the BiGRU layer captures sequential dependencies that complement the contextual embeddings generated by BERT, thereby enriching the model's temporal and contextual awareness. Although CapsNets were originally introduced for vision tasks to model spatial hierarchies, their underlying mechanism, capturing relationships between low-level and high-level features through dynamic routing, offers potential relevance for modeling compositional semantics in text. In Arabic discourse, expressions of optimism and pessimism often arise from context-dependent constructions, where sentiment emerges through the interaction of multiple linguistic elements, such as modifiers, negation, and implicit cues. In this work, the CapsNet component is incorporated to explore its ability to preserve such structured relationships within contextual and sequential representations. However, we acknowledge that the application of CapsNet in this context is not yet theoretically standardized. Therefore, we frame this component as an exploratory architectural extension rather than a definitively superior alternative. Nonetheless, developing specialized models tailored for optimism and pessimism detection remains essential, particularly for languages with unique linguistic characteristics such as Arabic.

The main contributions of this paper are as follows:

1. Introduces the Saudi optimism and pessimism detection (SOPD) dataset, the first and largest high-quality dataset specifically designed for optimism and pessimism detection in the Arabic language, with a particular focus on the Saudi dialect;
2. Establishes the first benchmark for future research on optimism and pessimism detection in Arabic;
3. Systematically evaluates the performance of seven state-of-the-art Arabic language models (Arabic-BERT, CamelBERT-DA, ARBERTv2, AraBERTv02, QARiB, MARBERTv2, and SaudiBERT) on the SOPD dataset, demonstrating their effectiveness in handling optimism and pessimism classification; and
4. Proposes a novel hybrid approach for optimism and pessimism detection using BERT-BiGRU-CapsNet.

These contributions enable rigorous comparative analyses and foster methodological advancements, thereby enhancing the understanding of nuanced sentiment detection in Arabic. Moreover, the classification of optimism and pessimism supports the promotion of individual well-being and societal health, while also offering valuable insights for the formulation of tailored mental health strategies. In addition, this work might enable businesses to automatically classify customer posts as optimistic or pessimistic, assisting marketing managers in analyzing customer reviews and providing deeper insights into customer behavior, engagement, and brand perception.

Related works

The inherent complexities of the Arabic language, stemming from its distinctive linguistic features and numerous dialects, pose substantial challenges for Arabic SA. These challenges are further exacerbated by the scarcity of resources, including adequately annotated datasets, comprehensive lexicons, and specialized programming tools^{9,10}. The rich morphological variation in Arabic further contributes to its linguistic complexity, requiring sophisticated approaches for accurate analysis. Moreover, research in Arabic SA remains limited compared to English.^{10–13}, primarily due to the lack of large-scale, high-quality annotated data. According to Alqahtani and Alothaim¹⁴, the size of existing Arabic datasets is extremely limited due to the fact that emotion detection in Arabic text is a relatively new research domain and there are only few datasets available for this task. Table 1 shows the recent work on emotion and hope dataset on Arabic. This deficit significantly hinders the development of accurate and robust models, particularly for low-resource languages like Arabic, where such data is essential for effective model training¹⁰. While most related studies focus on general emotion detection or Arabic SA^{15–23}, there remains a considerable gap in research dedicated specifically to optimism and pessimism detection. For example, Zaghouani and Biswas²³ proposed a joint emotion and hope-speech annotated dataset for English and Arabic. The dataset captures basic emotions such as joy, sadness, anger, love, and fear, along with their causes, as well as fine-grained hope-speech categories, including inspirational/motivational, solidarity/peace, resilience, and spiritual themes. It was constructed by filtering and recombining existing sources, resulting in 27,456 Arabic and 10,036 English instances. To classify the hope-speech categories, the authors benchmarked two traditional machine learning (ML) models (Logistic Regression and Naive Bayes) against the AraBERT transformer. AraBERT achieved an F1-Score of approximately 47% for Arabic hope-speech prediction.

AlDahoul and Zaki²² proposed multi-task, multi-modal models for detecting hope speech, hate speech/offensive content, and emotion in Arabic text. The authors evaluated the performance of base large language models (LLMs), fine-tuned LLMs, and pre-trained embedding models on the MAHED 2025 datasets, which included three subtasks: (1) classifying Arabic text into hope, hate, or not applicable; (2) emotion detection (12 classes), offensive language detection, and hate versus non-hate when offensive; and (3) detecting hateful memes by combining Arabic text and images. For modeling, they used fine-tuned LLMs (e.g., GPT-4o-mini, Gemini Flash 2.5), embedding models with traditional classifiers (e.g., SVM), and ensemble methods. They achieved a macro F1-Score of up to 72.1% on the hope/hate/not task.

Tăbușcă et al.²⁹ investigated the computational modeling of dispositional sentiment, focusing on optimism and pessimism in social media text. They proposed the RoBERTa-OPT-BinaryLabels model, which used interpretability techniques to highlight important linguistic patterns for each sentiment. The model was trained on the OPT dataset and achieved an accuracy of 83%. Optimistic instances were marked by tokens such as amazing, excited, happy and awesome, often including cultural references and celebrity names as indirect affective cues. In contrast, pessimistic predictions were dominated by emotionally intense and profane language, with words such as useless, and hate. The study demonstrated that transformer-based models, combined with interpretability techniques and psycholinguistic profiling, provided valuable insights into the affective and lexical characteristics of optimism and pessimism in social media text.

Almars¹³ proposed a deep learning model that combined an attention mechanism with a bidirectional long short-term memory (Bi-LSTM) network to classify depression in Arabic text. The model focused on discriminative features and learned significant word weights that strongly contributed to depression detection. It was trained on 6,000 collected tweets and achieved an accuracy of 83%. On the other hand, Mansy et al.³⁰ proposed an ensemble model combining three architectures: Bi-LSTM, Bi-GRU, and MARBERT, to address emotion detection in Arabic tweets. Two of these models, Bi-LSTM and Bi-GRU, are specialized recurrent neural networks (RNNs), while MARBERT is a pre-trained language model based on the BERT transformer. The authors evaluated their approach using the SemEval-2018-Task1-Ar-Ec dataset. Experiments demonstrated that MARBERT achieved superior results, improving macro F1-Score by 4.2% and micro F1-Score by 3.5%. Furthermore, the proposed ensemble model outperformed the individual models, exhibiting improvements ranging from 5.3 to 23.3% in macro F1-Score.

Dataset	Year	Language	# Samples	Domains / Source	Labels
IAEDS ²⁴	2019	Arabic (Iraqi)	About 1,300	Facebook posts	anger; disgust; fear; happiness; sadness and surprise
SemEval-2019 SenWave ²⁵	2020	Arabic/English	1,000 (Arabic)	Twitter and Weibo messages; COVID-19	optimistic; thankful; empathetic; pessimistic; anxious; sad; annoyed; etc.
AraEmoCorpus ²⁶	2021	Arabic	5.5 million	X/Twitter; COVID-19	anger; disgus; fear; joy; sadness; surprise and symptom or non-symptom
AEELEX (bilingual emotion lexicon) ²⁷	2021	Arabic	35,383	X/Twitter	anger; sadness; joy; surprise; disgust; trust; and anticipation
EmoHopeSpeech ²³	2025	Arabic/English	27,456 (Arabic)	X/Twitter; mixed sources (poetry, emotional tone)	emotions (happiness; sadness; love; joy; and others), emotion intensity (low; medium; medium; high and N/A) , complexity (simple; medium; complex and N/A), cause, and hope levels
Mulilingual-Hope-Speech ²⁸	2025	Arabic, English, and Kurdish	9217 (Arabic)	Instagram comments	hope; not_hope
MAHED shared Subtask-1	2025	Arabic	9,843	Social media;	hope; hate and not applicable
SOPD (ours)	2025	Arabic (Saudi dialect)	16,152	Finance; Sports; Hawamer & Kooora forums	optimistic; pessimistic; neutral

Table 1. Arabic emotion and hope-speech datasets.

Hamed et al.³¹ proposed a dialectal Arabic text emotion classification architecture that categorized expressions into four emotions: anger, joy, fear, and sadness. The authors evaluated ten versions of BERT models and proposed an ensemble model using a majority voting technique. This approach combined the three best-performing pre-trained BERT models: BERT-base-Arabertv2, BERT-medium-Arabic, and BERT-large-Arabertv02-Twitter into a single ensemble. The results were compared against eight traditional machine learning classifiers, such as Support Vector Machines, Naive Bayes, and Decision Trees, as well as the ten individual BERT models. Experiments were conducted on the Arabic tweets dataset for the EI-OC task, provided by SemEval, which contained 5,600 tweets. The proposed ensemble approach achieved an accuracy of approximately 84%, while the highest accuracy among the other investigated models was 76%.

Shahriar et al.³² explored deep learning methods to classify Arabic poems into emotional categories. The authors utilized four models: one-dimensional Convolutional Neural Networks (1D-CNNs), bidirectional RNNs, CNN-LSTM, and AraBERT, and compared their performance against five traditional machine learning classifiers and three ensemble methods. The ML classifiers included SVM, k-Nearest Neighbors, Multi-Layer Perceptrons, Decision Trees, and Gaussian Naïve Bayes, while the ensemble methods comprised Gradient Boosting, Adaptive Boosting, and Random Forest. The authors introduced a new dataset of 9,452 Arabic poems labeled with the emotional categories of joy, sadness, and love, collected from an Arabic poetry website. The findings indicated that traditional deep learning models, specifically 1D-CNNs, Gated Recurrent Units (GRUs), and LSTMs, achieved F1-Scores of 62%, 62%, and 53%, respectively. In contrast, the AraBERT model demonstrated superior performance, attaining an F1-Score of 77%.

Al-Maamari et al.³³ explored a hybrid deep learning approach to classify both printed and handwritten Arabic characters. The authors proposed a hybrid transformer encoder-based model that integrates transfer learning with self-attention mechanisms. The methodology utilized two pre-trained CNNs, VGG16 and ResNet50, as feature extractors, followed by a feature ensemble process. These extracted features were then processed by a Transformer encoder architecture, which leverages self-attention and Multilayer Perceptron components to capture global dependencies and refine character representations. The authors evaluated their model using two primary datasets: the Arabic OCR dataset for printed text and the Arabic Handwritten Character Recognition (AHCR) dataset. The performance of this hybrid architecture was compared against individual CNN-based models and traditional ensemble techniques. The findings indicated that the proposed model achieved exceptional results, attaining an accuracy of 99.51% for printed characters and 98.19% for handwritten characters. These results demonstrated that the integration of Transformer-based global context with CNN-based spatial features outperforms standalone architectures in handling the complexities of the Arabic script.

Mittal et al.³⁴ examined hate speech detection using deep learning and transformer-based approaches. The study addressed a three-class classification task (hate, offensive, and non-hate). The results showed that transformer-based models, particularly fine-tuned RoBERTa, significantly outperformed CNN and BiLSTM architectures, demonstrating the effectiveness of contextualized language representations for hate speech detection. Karaouglan and Findik³⁵ proposed an unsupervised rule-based method for opinion target extraction that integrates novel preprocessing, extended syntactic rules, and majority voting. Tested on English restaurant reviews, it outperformed unsupervised baselines and achieved performance comparable to supervised methods.

Blanco and Lourenço³⁶ combined BERT embeddings with BiLSTM networks to analyze expressions of optimism and pessimism in Twitter conversations about COVID-19. The study systematically compared multiple neural architectures, including CNNs and hybrid models, while employing two distinct text embedding approaches (GloVe and BERT) across three different network types (linear networks, CNNs, and BiLSTMs). The experimental evaluation used a dataset of 150,503 public tweets. Results showed that the BERT-BiLSTM architecture achieved superior performance in emotion detection. In addition, the authors contributed two novel annotated Twitter corpora specifically designed for optimism and pessimism detection. The analysis revealed that conversations with strong pessimistic signals showed little emotional change (62.21%), whereas only 10.42% of optimistic conversations maintained their initial mood. The study further identified distinct patterns of emotional volatility: strongly pessimistic conversations exhibited minimal variation but revealed significant correlations between emotional fluctuations and social influence dynamics.

Al-Laith and Alenezi²⁶ investigated the classification of emotions and COVID-19 symptoms in Arabic tweets. The authors evaluated two deep learning architectures, namely LSTM and CNN, on a dataset comprising approximately 5.5 million collected tweets. Using a rule-based lexicon approach for automated labeling, the study categorized the content into six emotion classes and identified the presence of physical symptoms. The results demonstrated that the LSTM model outperformed the CNN, achieving an average F1-Score of 82% for emotion detection and 75% for symptom classification. Furthermore, the findings revealed that joy constituted the most frequently expressed emotion in the dataset. However, tweets associated with anger and fear generated higher levels of user engagement, and retweets were predominantly used to disseminate symptom-related information.

Almerekhi et al.²⁷ investigated the influence of geographical and cultural contexts on the emotional expression of X/Twitter users by comparing a Holy City (Makkah) with a non-pilgrimage city (Riyadh) in Saudi Arabia. The authors developed a bilingual emotion lexicon named AEELex (Arabic English Emotion Lexicon), designed specifically to capture nuanced emotional indicators in both MSA and the Saudi dialect. The study utilized a dataset of 35,383 geolocated tweets collected from the two target cities. The study evaluated multiple classifiers, including SVM and Naive Bayes. The authors classified content into Plutchik's eight basic emotion categories (fear, anger, sadness, joy, surprise, disgust, trust, and anticipation). The experimental results indicated that the SVM model achieved the best performance in capturing local dialectal nuances. The analysis revealed distinct emotional patterns across the two cities: while joy and trust were common in both, Makkah showed a higher concentration of spiritually oriented positive emotions, whereas Riyadh exhibited greater emotional variability, with elevated levels of anticipation and anger linked to socio-economic pressures.

Alshahrani et al.³⁷ proposed a predictive model for classifying optimism and pessimism in Twitter messages. The authors fine-tuned XLNet models to classify sentiments at both the individual message level and the user level. The methodology involved training multiple XLNet-based models using extra-long transformers. Three datasets were employed: the optimism–pessimism Twitter dataset (OPT) published by Ruan et al.³⁸, an emotional dataset published by Saravia et al.³⁹, and the Twitter Sentiment Analysis dataset (TSA). In total, the combined dataset consisted of 1.9M English tweets. The results demonstrated that the fine-tuned XLNet models achieved high classification accuracy, with the base and large models reaching up to 97.92%, and averaging 96.16% and 96.45%, respectively, while reducing the error rate by 54.24%. Additionally, the authors observed that optimistic messages predominantly contained positive emotions and sentiments, whereas pessimistic ones were characterized by negative emotions. Alshahrani⁴⁰ also fine-tuned a BERT model for the same task and introduced Soft Label Assignment (SLA), a method for assigning probabilities to each class. Two datasets were used: the OPT dataset and an emotional dataset. The fine-tuned BERT model achieved an accuracy of up to 100% at the user level and 97.10% at the tweet level on the test set, excluding neutral messages.

Ruan et al.³⁸ investigated the classification of optimistic and pessimistic tweets using a bag-of-words representation. The authors evaluated six ML classifiers: Naïve Bayes, Nearest Neighbor, Decision Tree, Random Forest, Gradient Boosting, and Stochastic Gradient Descent on the OPT dataset. The results demonstrated that the Gradient Boosting classifier outperformed the others. However, since the bag-of-words approach was used to represent tweets, the method lacked the contextual understanding inherent in more advanced models, which limited its accuracy. Subsequently, Caragea et al.⁴¹ employed three DL models CNNs, BiLSTMs, and RNNs using GloVe embeddings⁴². Two datasets were used: OPT and TSA. Their approach achieved substantially higher performance compared to Ruan et al.³⁸. However, a key limitation was the use of static GloVe embeddings, which cannot adapt to contextual variations. This limitation reduced the models' effectiveness in handling polysemous words (words with multiple meanings) and syntactic structures such as negation, ultimately constraining their ability to capture nuanced linguistic and contextual details.

Compared to SA and emotion studies, optimism and pessimism detection has received limited attention. Moreover, while some studies have explored emotions in Arabic, no prior work has specifically focused on detecting optimism and pessimism in the Arabic language. Therefore, this paper exclusively investigates the detection of optimism and pessimism in Arabic. We evaluate the performance of seven advanced Arabic language models (Arabic-BERT, CamelBERT-DA, ARBERTv2, AraBERTv02, QARiB, MARBERTv2, and SaudiBERT) on the collected dataset. In addition, we propose a hybrid architecture, BERT-BiGRU-CapsNet, which combines a BiGRU layer with a CapsNet on top of a BERT encoder for optimism and pessimism classification.

The proposed hybrid architecture

This section presents a detailed description of the proposed hybrid architecture. The objective of this study is to investigate the performance of various Arabic language models in detecting optimism and pessimism. We propose a hybrid architecture that combines a Bidirectional GRU (BiGRU) layer⁴³ with a Capsule Network (CapsNet)⁴⁴ on top of a BERT encoder. Figure 1 illustrates the overall architecture, which is summarized as follows:

1. **BERT Encoding:** A pretrained BERT model (SaudiBERT) is used to obtain contextual embeddings for the input text.
2. **BiGRU Layer:** The embeddings from BERT are passed to a Bidirectional GRU to capture sequential dependencies.
3. **Capsule Layer:** The final hidden representation from the GRU layer is mapped into capsule vectors and passed through a squash activation function.

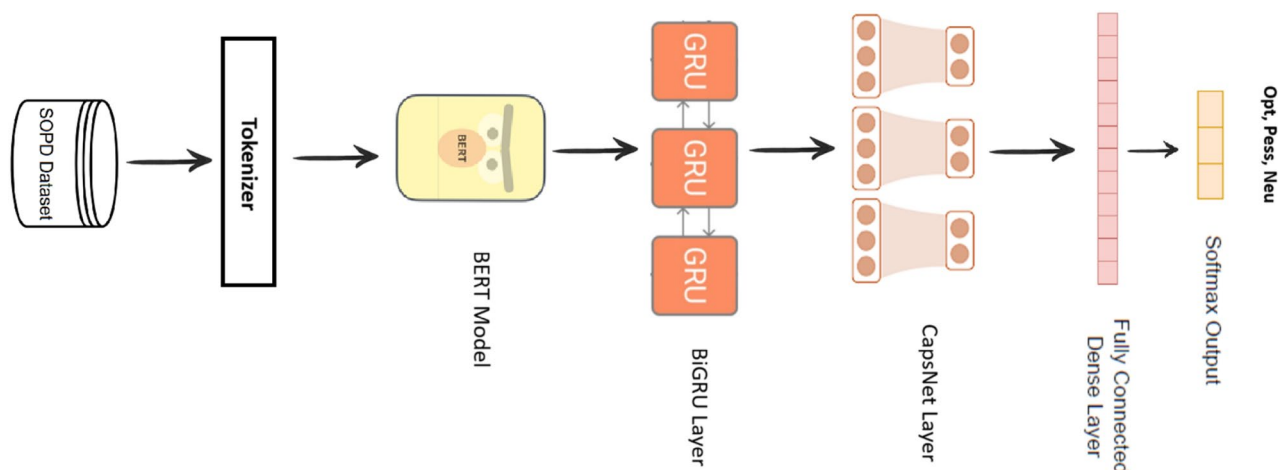


Fig. 1. Overview of the proposed SaudiBERT-BiGRU-CapsNet hybrid model architecture.

4. *Classification*: The capsule vectors are flattened and fed into a linear layer for optimism and pessimism classification.

BERT encoder

Given an input text sequence

$$\mathbf{x} = \{x_1, x_2, \dots, x_T\} \quad (1)$$

of length T , the text sequence is first tokenized,

$$\mathbf{x}_{\text{tok}} = \text{Tokenizer}(\mathbf{x}_{\text{raw}}). \quad (2)$$

The tokenized sequence is then fed to the BERT model, which produces the last hidden state $\mathbf{H}$ of shape (B, T, d) , where B is the batch size and d is BERT's hidden dimension (typically $d = 768$). Formally,

$$\mathbf{H} = \text{BERT}(\mathbf{x}), \quad \mathbf{H} \in \mathbb{R}^{B \times T \times d}. \quad (3)$$

Here, $\mathbb{R}^{B \times T \times d}$ denotes a 3D tensor of real numbers with shape (B, T, d) .

Here, $\mathbf{H}$ captures rich contextual information for each token in the sequence, utilizing self-attention mechanisms at multiple layers.

Bidirectional GRU layer

After that, $\mathbf{H}$ is fed into a single-layer bidirectional GRU (BiGRU) with hidden dimension $h = 128$. For a single GRU cell, the hidden state update at time step t can be written as:

$$\mathbf{z}_t = \sigma\left(\mathbf{W}_z \begin{bmatrix} \mathbf{h}_{t-1} \\ \mathbf{x}_t \end{bmatrix} + \mathbf{b}_z\right), \quad (4)$$

$$\mathbf{r}_t = \sigma\left(\mathbf{W}_r \begin{bmatrix} \mathbf{h}_{t-1} \\ \mathbf{x}_t \end{bmatrix} + \mathbf{b}_r\right), \quad (5)$$

$$\tilde{\mathbf{h}}_t = \tanh\left(\mathbf{W} \begin{bmatrix} \mathbf{r}_t \odot \mathbf{h}_{t-1} \\ \mathbf{x}_t \end{bmatrix} + \mathbf{b}\right), \quad (6)$$

$$\mathbf{h}_t = (1 - \mathbf{z}_t) \odot \mathbf{h}_{t-1} + \mathbf{z}_t \odot \tilde{\mathbf{h}}_t, \quad (7)$$

where $\mathbf{z}_t$ and $\mathbf{r}_t$ are the update and reset gates, σ is the sigmoid function, and $\odot$ denotes element-wise multiplication.

For bidirectional GRUs, the sequence is processed in both forward and backward directions, and the outputs are concatenated. We denote the final hidden states by $\mathbf{h}_N$. For a single-layer BiGRU:

$$\mathbf{H}_{\text{GRU}}, \mathbf{h}_N = \text{BiGRU}(\mathbf{H}), \quad (8)$$

where $\mathbf{H}_{\text{GRU}} \in \mathbb{R}^{B \times T \times 2h}$ is the full sequence of GRU outputs and $\mathbf{h}_N$ contains the final hidden states in both directions (shape $[2, B, h]$ for a single-layer GRU). We then concatenate the final forward and backward hidden states $\mathbf{h}_N^{(\text{fw})}$ and $\mathbf{h}_N^{(\text{bw})}$:

$$\mathbf{h}_{\text{final}} = \begin{bmatrix} \mathbf{h}_N^{(\text{fw})} \\ \mathbf{h}_N^{(\text{bw})} \end{bmatrix} \in \mathbb{R}^{B \times (2h)}. \quad (9)$$

In the proposed model, a single BiGRU layer has been used with $h = 128$ and a dropout probability of 0.1.

Capsule layer

The final hidden state $\mathbf{h}_{\text{final}}$ of the BiGRU layer is then mapped into a set of capsule vectors. Let C be the number of capsules, each of dimension D . In the proposed model, $C = 10$ and $D = 16$. First, a linear transformation is applied:

$$\mathbf{u} = \mathbf{W}_{\text{caps}} \mathbf{h}_{\text{final}} + \mathbf{b}_{\text{caps}}, \quad \mathbf{u} \in \mathbb{R}^{B \times (C \cdot D)}, \quad (10)$$

where $\mathbf{W}$ and $\mathbf{b}$ are the trainable weight matrix and bias vector of the capsule layer.

The projected output $\mathbf{u}$ (or logits for the capsule layer) is then reshaped into $\mathbf{U} \in \mathbb{R}^{B \times C \times D}$. Next, the *squash* activation function is applied to each capsule vector. Following the original CapsNet formulation, the squash function is defined as:

$$\text{squash}(\mathbf{u}) = \frac{\|\mathbf{u}\|^2}{1 + \|\mathbf{u}\|^2} \frac{\mathbf{u}}{\|\mathbf{u}\| + \epsilon}, \quad (11)$$

where $\|\mathbf{u}\|$ denotes the ℓ_2 -norm of $\mathbf{u}$, and ϵ is a small constant (e.g., 10^{-8}) to avoid division by zero. Applying this function to each capsule vector yields the final capsule representation $\mathbf{V} \in \mathbb{R}^{B \times C \times D}$.

Classification layer

Following the Capsule Layer, we obtain a set of capsule vectors $V \in \mathbb{R}^{B \times C \times D}$. To produce the final prediction, these capsule outputs are first *flattened* into a single vector per sample:

$$z = \text{reshape}(V), \quad z \in \mathbb{R}^{B \times (C \cdot D)}. \quad (12)$$

Then z is fed into a fully connected layer, mapping $(C \cdot D)$ dimensions to 3 classes:

$$\text{logits} = W_{\text{clf}} z + b_{\text{clf}} \in \mathbb{R}^{B \times 3}. \quad (13)$$

Here, the three output classes are *optimistic*, *pessimistic*, and *neutral*. The softmax function is applied to the logits and a standard cross-entropy loss is used for training:

$$\mathcal{L} = - \sum_{i=1}^B \sum_{k=1}^3 y_{ik} \log(\text{softmax}(\text{logits}_i)_k), \quad (14)$$

where y_{ik} is the ground-truth label for sample i and class k , and $\text{softmax}(\cdot)$ is the softmax function. This setup enables the model to classify each sample as *optimistic*, *pessimistic*, or *neutral* based on its learned representations.

Experimental setting

In this section, we outline the experimental setup employed in this study. The setup is organized into four main subsections: data collection, Arabic language models, model fine-tuning, and evaluation metrics. We describe the data collection process, including dataset creation, annotation procedures, and descriptive statistical analysis. Next, we provide a concise description of the Arabic language models used, along with the fine-tuning methodology. Finally, we specify the evaluation metrics applied to assess model performance. These steps are essential for ensuring the effectiveness of the models in detecting optimism and pessimism within the dataset.

Data collection

This section provides a brief overview of the creation of the dataset used in this study, referred to as the Saudi optimism and pessimism detection (SOPD) dataset. SOPD is the first and largest Arabic dataset covering only optimism and pessimism traits. Figure 2 illustrates the SOPD dataset creation workflow, highlighting the stages of data collection, pre-processing, and manual annotation.

The workflow begins with data collection, where posts and replies were gathered from the Hawamir⁴⁵ and Kooora⁴⁶ forums, focusing on Arabic content related to finance and sports. Following data collection, a pre-processing step was applied, including the removal of URLs, user mentions, and entries containing three words or fewer. Finally, the dataset underwent manual annotation, where each entry was classified into one of three sentiment categories: optimistic, pessimistic, or neutral. This systematic process produced a well-structured dataset.

The dataset was collected from two prominent Saudi forums: Hawamir and Kooora. Hawamir is a comprehensive, interactive platform and one of the leading Arabic websites specializing in financial and investment discussions, with a particular focus on stock markets and the Saudi stock exchange. The site features

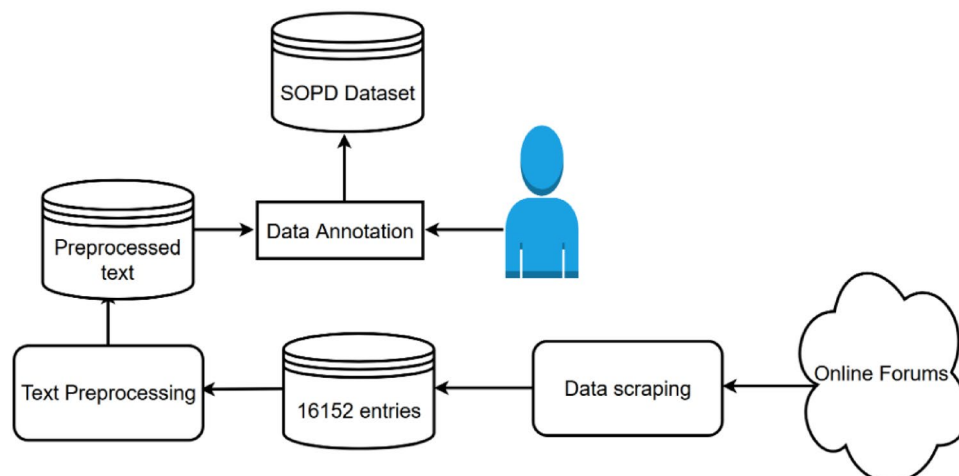


Fig. 2. SOPD dataset creation workflow, presenting the stages of data collection, preprocessing (including deduplication and filtering of non-relevant and identifying content), and manual annotation for optimism–pessimism detection. The reported total of 16,152 samples reflects the final post-filtering annotated corpus, on which a stratified 80–20 train/test split was subsequently applied.

various content tailored to investors and individuals interested in financial markets, including active discussion forums where members exchange views on stock movements, market predictions, and corporate financial announcements. Kooora, on the other hand, is a widely recognized Arabic sports forum, primarily focused on football, and is among the largest sports platforms in the Arab world. It offers dedicated sections for discussing both local and international football, including European leagues, cups, and global tournaments. The forum also covers other sports such as basketball, tennis and athletics, in addition to general topics. It provides up-to-date sports news that users can discuss and analyze. The platform hosts a vibrant community where members share opinions, predictions, and analyses of sports events, as well as engage in competitions, general discussions, and social topics. The data, written in Arabic, was collected between November 17, 2024, and January 11, 2025, using the BeautifulSoup API in Python for web scraping. The data collection process was relatively straightforward due to the simple structure of the target websites and the high quality of the text content available on these platforms. This simplicity reduced the need for complex preprocessing or elaborate collection mechanisms. Only limited text preprocessing was applied, which involved removing URLs, user mentions, and entries containing three words or fewer.

Annotation was conducted by the second author, who is a native speaker of Saudi Arabic with expertise in SA and dialectal Arabic NLP, ensuring that domain-specific and linguistic nuances were consistently captured for optimism detection across three sentiment categories: optimism, pessimism, and neutral. Each text sample was assigned a single label. The data was collected from Saudi online forums, where the language is predominantly in Saudi dialectal Arabic, often including colloquial expressions, informal writing styles, and emphasis markers. The following guidelines were established to ensure annotation consistency and reliability. In terms of general rules, every sample was required to receive exactly one label corresponding to optimism, pessimism, or neutral. The assigned annotation reflected the overall tone of the text rather than isolated words. When a text contained mixed sentiments, the stronger or more dominant tone determined the label. Dialectal expressions, idioms, and punctuation were also considered in order to capture sentiment strength more accurately. The three labels were defined as follows. Optimism was used when the text expressed positive expectations, hope, or confidence in future outcomes, often including encouragement, success, or improvement. Pessimism was assigned when the text conveyed negativity, disappointment, hopelessness, fear, or negative expectations about the future, including sarcastic or cynical remarks with a clearly negative undertone. Neutral was applied to texts that did not clearly reflect optimism or pessimism, such as factual or descriptive statements without strong emotional cues.

Certain edge cases required specific treatment. When both optimism and pessimism appeared in the same text, the label of the stronger sentiment was assigned. For example, in the sentence “The situation is difficult, but we have great hope”, the dominant tone is optimistic, so the label was as optimism. In the case of sarcasm or irony, the label reflected the underlying sentiment rather than the surface meaning. For instance, “Sure, everything will improve, just like every time!” is sarcastic and therefore it was labeled as pessimistic. Punctuation was also considered: strong positive emphasis suggested optimism, while negative emphasis markers or repetition of negative words indicated pessimism. Finally, when the text is purely descriptive or objective, such as “Today the market opened at 10”, or when optimism and pessimism are expressed with equal strength and no dominant sentiment can be identified, such as “The news today is both discouraging and encouraging at the same time”, the text was labeled as neutral. Several quality control measures were applied to strengthen reliability. A pilot annotation phase was conducted on a subset of the data to refine and finalize the guidelines. Randomly selected samples were later re-examined to ensure consistent application of rules. To further assess stability, a portion of the data was re-annotated by the same annotator after a time gap, allowing a self-consistency check. Ambiguous cases were recorded in an annotation log with detailed rationales, and the aforementioned guidelines were consistently applied to ensure transparency and traceability.

Table 2 presents the class distribution across the two domains: finance and sport. Each entry is annotated into three sentiment classes: optimistic (6,611), pessimistic (4,069), and neutral (5,472), resulting in a total of 16,152 annotated samples. The class imbalance, characterized by a higher proportion of optimistic samples compared to pessimistic ones, results in only a marginal difference in model performance. This is particularly evident because the dataset includes a sufficient number of instances to represent each class adequately. Overall, the imbalance has negligible effects on the results, as demonstrated in the Results section. Under macro-averaged evaluation and multi-seed experimentation, no substantial imbalance-induced performance degradation was observed. Nevertheless, incorporating a greater number of pessimistic samples in future work will be essential to enhance dataset balance and improve overall robustness. Furthermore, the predominance of optimistic sentiment among users likely reflects the inherent nature of these platforms, particularly within the financial domain. This predominance may also be influenced by Islamic cultural values, given that optimism is a fundamental aspect of Arab cultural identity⁴⁷.

Table 3 presents a sample from the SOPD dataset, illustrating a subset of entries across both domains and categories. Each entry is annotated with its respective sentiment label (optimistic, pessimistic, or neutral), which

Domain	Optimistic	Pessimistic	Neutral
Finance	4130	2207	2834
Sport	2481	1862	2638
Total	6611	4069	5472

Table 2. Class distribution across domains.

Class	Sentiment label	Original text	Translation
Sport	Pessimistic	إذا ما نفوز ب 5 هذا ماهو فوز !	If we don't win by 5, it's not a win!
Sport	Optimistic	اسهل فريق ممكن تسجل فيه هو النصر	The easiest team to score against is Al-Nasr.
Sport	Neutral	ياشباب النصر لعب في الدوري 13 مباراة سجل عليه 15 هدف	Oh youth, Al-Nasr played 13 matches in the league and scored 15 goals.
Finance	Pessimistic	للأسف نظامهم عطلا. خدمة التداول عندهم 0 من 100	Unfortunately, their system is out of order. Their trading service is 0 out of 100.
Finance	Neutral	القفل القديم ما عاد يفتح	The old lock no longer opens
Finance	Optimistic	دامها لصندوق والله ما ابيع سهم لين تدبل كم مره	As long as it is for a fund, I swear I will not sell the share until it doubles several times.

Table 3. Sample of SOPD dataset.

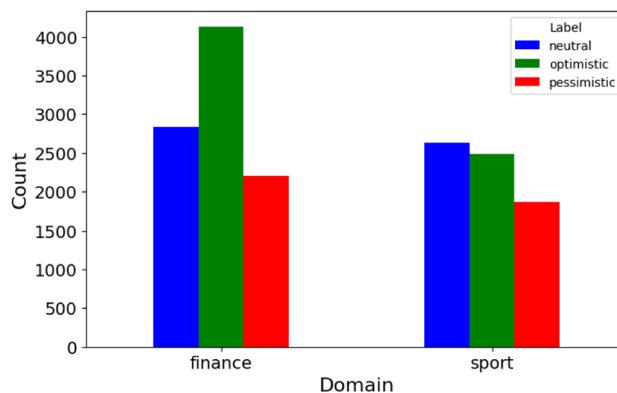


Fig. 3. Class distribution across domains in the SOPD dataset: Finance (Optimistic = 4130, Pessimistic = 2207, Neutral = 2834) and Sport (Optimistic = 2481, Pessimistic = 1862, Neutral = 2638), with overall totals of 6611 Optimistic, 4069 Pessimistic, and 5472 Neutral instances.

demonstrates the dataset's organization and sentiment classification. In addition, each entry is linked to one of two domains, sport or finance, along with the corresponding textual data.

Figure 3 presents the class distribution of sentiment labels across two domains in the SOPD dataset: finance and sport. In finance, the optimistic class (≈ 4000) dominates, followed by neutral (≈ 2800) and pessimistic (≈ 2000). This suggests a prevalence of positive sentiment in financial texts. In sport, neutral (≈ 2700) and optimistic (≈ 2500) are nearly equal, while pessimistic (≈ 1800) is the least frequent. This indicates a more balanced sentiment in sports content.

Figure 4 illustrates the text length distribution in the SOPD dataset, showing the frequency of texts based on the number of words. Most texts contain 4–8 words, with a peak at 6 words (≈ 1500 occurrences), while 5- and 7-word texts each exceed 1,200 instances. A sharp decline is observed after 10 words, though texts of 10–14 words remain relatively frequent, ranging from ≈ 1000 to ≈ 900 occurrences. Longer texts (17–20 words) are rare, with fewer than 700 occurrences.

Arabic language models

In this section, we briefly describe all the Arabic language models used in the experiment. Table 4 presents a summary of the key characteristics of the Arabic pre-trained models. The evaluation process was conducted by measuring the performance of the following Arabic language models:

- *Arabic-BERT*: pretrained on about 8.2 billion words from Arabic OSCAR, Wikipedia, and other sources, totaling approximately 95 GB of text⁴⁸. It was trained for 3M steps with a batch size of 128, retaining some non-Arabic words for tasks such as named entity recognition, and includes both MSA and Dialectal Arabic (DA).
- *CamelBERT-DA*: a variant of CAMELBERT, a transformer-based Arabic language model designed to improve NLP tasks by capturing the complexities of both MSA and dialectal variations⁴⁹. CamelBERT-DA was pre-trained on a dataset composed of 54 GB of DA text compiled exclusively for dialectal Arabic. The model was trained for the first 90,000 steps with a batch size of 1,024, and then continued with a batch size of 256 for the remainder. It employed whole-word masking with a duplicate factor of 10, using a maximum sequence length of 128 tokens, except for one dataset with a 512-token limit, where the maximum length was 80.

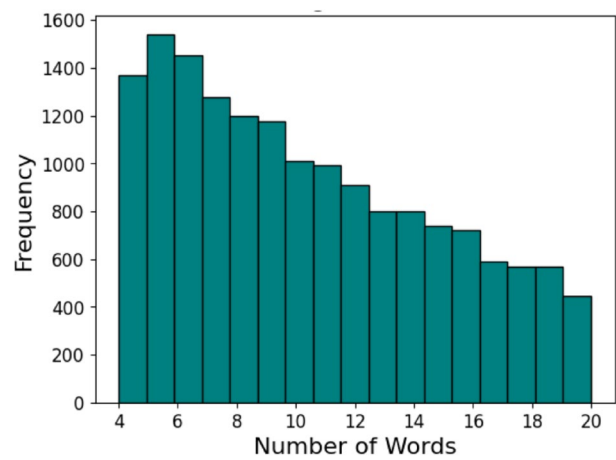


Fig. 4. Text length distribution of the SOPD dataset (Median: 8 words; 95th Percentile: 18 words). Note: The 95th percentile indicates that 95% of the corpus is captured within a 18 words.

Model	#Para	Tokenizer	Vocab size	Dataset size	Dataset type	Dataset sources
Arabic-BERT	110M	WordPiece	32K	95GB	MSA	OSCAR, Arabic Wikipedia, OPUS, etc.
CamelBERT-DA	108M	WordPiece	30K	54GB	DA	28 different Arabic dialectal corpora
ARBERTv2	110M	WordPiece	32K	95GB	MSA	OSCAR, Arabic Wikipedia, OPUS, etc.
AraBERTv02	135M	WordPiece	64K	70GB	MSA+DA	Arabic Gigaword, OSIAN, OPUS, Tweets
MARBERTv2	163M	WordPiece	100K	198GB	MSA+DA	MARBERTv1 + web and social media data
QARiB	136M	WordPiece	64K	97GB	MSA+DA	Gigaword 5, OBUS, El-Khair, 420M tweets
SaudiBERT	144M	SentencePiece	75K	26.3GB	Saudi DA	STMC + SFC

Table 4. Key characteristics of the Arabic Pre-trained models employed in this experiment.

- *ARBERTv2*: the updated version of the ARBERT model. It was further pretrained on a dataset containing 243 GB of text and 27.8 billion tokens⁵⁰.
- *AraBERTv02*: an improved version of earlier iterations within the AraBERT family, a series of Arabic language models based on Google’s BERT architecture. These models were trained on extensive Arabic corpora, including Wikipedia, OSCAR, and news datasets⁵¹. The AraBERTv0.2-base model was trained on a TPuv3-8 processor with 420 million and 207 million examples for sequence lengths of 128 and 512, respectively. Training employed batch sizes of 2,560 for 1 million steps and 384 for 2 million steps, resulting in a total of 3 million training steps.
- *MARBERTv2*: an improved version of MARBERT, pretrained on the same multi-dialectal Arabic data (MSA and dialects) used in ARBERT and MARBERT, in addition to the AraNews dataset. It was trained with a larger sequence length of 512 tokens over 40 epochs on a corpus of approximately 29 billion tokens⁵².
- *QARiB*: a pretrained language model trained on about 420 million tweets and 180 million sentences from sources such as Arabic GigaWord, the Abulkhair Arabic Corpus, and OPUS⁵³. With a dataset of 14 billion tokens, a 64K vocabulary, 12 layers, and 10 million training iterations, QARiB is optimized for various NLP tasks in both MSA and DA.
- *SaudiBERT*: the first pretrained large language model focused exclusively on Saudi dialect text, trained on two large-scale corpora: the Saudi Tweets Mega corpus (+141M tweets) and the Saudi Forum corpus (+70M sentences), totaling 26.3 GB of text⁵⁴. It was trained with extended preprocessing and optimization, making it well suited for various NLP tasks involving Saudi dialects, such as sentiment analysis and text classification.

Models fine-tuning

This section describes the fine-tuning process for the models used in this experiment. The dataset was divided into training and test sets with an 80–20 split, where the training set contained 12,921 samples and the test set included 3,231 samples. To ensure robust evaluation, four different random seeds were used: 0, 1, 42, and 100. Each seed was run three times, resulting in a total of 12 experiments per model. The models were fine-tuned for 8 epochs with a batch size of 64, a learning rate of 5e-5, and a linear scheduling strategy that decayed the learning rate from its initial value to zero over the course of training. During training, the models were evaluated on the test set every 50 steps to monitor performance and ensure reliable assessment across runs.

Although k-fold cross-validation is often considered a rigorous evaluation strategy, in large-scale NLP tasks using pretrained language models such as BERT it is frequently computationally prohibitive. Instead, a widely adopted alternative is to evaluate performance using fixed train/test splits generated with single or multiple specific random seeds^{55–58}, which provide distinct yet reproducible experiments. However, prior work has

highlighted that language model performance can vary substantially across different splits, and an evaluation experiment based on a single dataset split using a specific seed may lead to unstable or non-generalizable conclusions^{59,60}. Therefore, the use of four seeds (0, 1, 42, 100) ensures stable, reproducible, and statistically sound evaluation while remaining computationally feasible.

Code availability

The dataset and the code files, along with the results, are available online^{61,62}. All experiments were conducted on a local machine with AMD Ryzen-9 7950x processor, 64GB DDR5 memory, and two GeForce RTX 4090 GPUs with 24GB memory each. The software environment was set up on the Ubuntu 22.04 operating system, and we used the HuggingFace Transformers library⁶³ in pre-training and fine-tuning the models, in addition to downloading many of the datasets used in the experiments from the HuggingFace hub.

Evaluation metrics

The selection of appropriate evaluation metrics is crucial for accurately assessing the performance of the proposed model. A range of metrics commonly employed in multi-class classification tasks was considered, as established in the literature. In this study, the model's performance was evaluated using Accuracy (15), Precision (16), Recall (17), and macro F1-Score (18). Accuracy measures the overall proportion of correctly classified samples, providing a straightforward indicator of general performance. Precision reflects the extent to which positive predictions are correct, while Recall indicates the proportion of actual positives that were successfully identified. The F1-Score provides a single measure of classification performance, particularly valuable when both false positives and false negatives must be considered. Given the imbalance among dataset classes, the macro F1-Score was employed, which averages the F1 values across classes and assigns equal weight to each, regardless of class size. As recommended by Gargiulo et al.⁶⁴, macro-F1 is particularly suitable for multi-class problems with class imbalance, as it ensures consistent evaluation across all categories.

These metrics were selected because they are commonly employed^{65–68}, allowing a comprehensive assessment of a model's effectiveness in capturing relevant patterns and predicting the sentiment polarity of optimism and pessimism within posts and comments.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$
 (15)

$$Precision = \frac{TP}{TP + FP}$$
 (16)

$$Recall = \frac{TP}{TP + FN}$$
 (17)

$$F1-Score = \frac{2 \cdot TP}{2 \cdot TP + FP + FN}$$
 (18)

Here, *TP* denotes true positives, *TN* true negatives, *FP* false positives, and *FN* false negatives.

Results

This section reports the experimental results obtained from evaluating multiple state-of-the-art Arabic language models alongside the proposed hybrid approach for classifying optimism and pessimism. Table 5 summarizes the performance metrics, with the primary focus on the F1-Score, which served as the benchmark for model efficacy. Notably, SaudiBERT achieved the highest F1-Score of 93.11%, thereby demonstrating exceptional capability in distinguishing between optimistic and pessimistic sentiments. MARBERTv2 followed with an F1-Score of 87.25%, exhibiting competitive performance that, while robust, was slightly inferior to that of SaudiBERT. QARIB also yielded a strong performance with an F1-Score of 86.39%, further reinforcing its classification reliability. In contrast, ARBERTv2 and CamelBERT-DA attained moderate performance levels with F1-Scores of 82.78% and 81.29%, respectively, whereas Arabic-BERT recorded the lowest performance, with an F1-Score of 79.88%. The differences in performance highlighted the effect of various pretraining methods and design choices, with SaudiBERT proving to be the most effective model for generalizing across the dataset.

Model	Accuracy	Precision	Recall	F1-Score
Arabic-BERT	80.56%	79.70%	80.16%	79.88%
CamelBERT-DA	82.20%	81.22%	81.38%	81.29%
ARBERTv2	83.57%	82.83%	82.74%	82.78%
AraBERTv02	85.42%	84.70%	85.12%	84.85%
QARiB	87.03%	86.36%	86.42%	86.39%
MARBERTv2	87.71%	87.30%	87.20%	87.25%
SaudiBERT	93.38%	93.16%	93.07%	93.11%
SaudiBERT-BiGRU-CapsNet	93.56%	93.38%	93.31%	93.34%

Table 5. Evaluation of Arabic language models on the SOPD dataset for optimism and pessimism detection.

Metric	F-statistic	p-value
Accuracy	495.46	2.30×10^{-24}
F1-score	501.94	1.97×10^{-24}

Table 6. ANOVA results for accuracy and F1-Score across all models on the validation set.

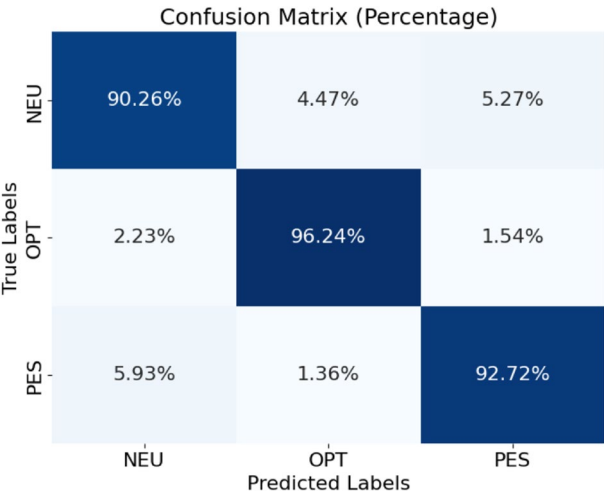


Fig. 5. Confusion matrix illustrating the predictions of the hybrid model SaudiBERT-BiGRU-CapsNet for each class on the validation set for the optimism and pessimism detection task.

Furthermore, the proposed hybrid model, SaudiBERT-BiGRU-CapsNet, outperformed all baseline models by achieving an F1-Score of 93.34%. This result suggested that the hybrid architecture successfully balanced the precision-recall trade-off. In this model, SaudiBERT captured deep semantic features, BiGRU modeled sequential dependencies, and CapsNet hierarchically organized spatial and relational patterns. Consequently, this integrated approach enhanced the model’s capacity to discern nuanced optimism and pessimism cues while effectively mitigating issues related to class imbalance.

Table 6 presents the one-way ANOVA results for accuracy and F1-Score across all models on the validation set. A one-way ANOVA is conducted on both accuracy and F1-Score using four random data splits (seeds 0, 1, 42, and 100) to evaluate the statistical significance of performance differences among the models. The analysis revealed a significant effect of the model type on accuracy ($F(7, 24) = 495.46, p < 0.001$) and F1-Score ($F(7, 24) = 501.94, p < 0.001$). These results confirm that the observed differences in model performance are not due to random variation but are statistically significant.

Figure 5 demonstrates the classification performance of the SaudiBERT-BiGRU-CapsNet model across three sentiment categories: neutral (NEU), optimistic (OPT), and pessimistic (PES). The model exhibited high accuracy, correctly classifying 90.26% of neutral instances, 96.24% of optimistic instances, and 92.72% of pessimistic instances. Despite this strong performance, some misclassification patterns were observed. Specifically, 4.47% of neutral texts were misclassified as optimistic, while 5.27% were predicted as pessimistic, indicating that certain neutral statements may contain subtle sentiment markers. Optimistic texts were the most accurately identified, with only 2.23% misclassified as neutral and 1.54% as pessimistic, suggesting that optimism was the most distinguishable sentiment category. In contrast, 5.93% of pessimistic samples were misclassified as neutral, and 1.36% were mistakenly labeled as optimistic, implying that some negative expressions may lack strong pessimistic cues. These results highlighted the model’s robustness while also pointing to challenges in distinguishing neutral statements from sentiment-laden ones, which could be addressed through further refinements in model training and dataset augmentation.

Figure 6 presents the training metrics of the proposed hybrid model, SaudiBERT-BiGRU-CapsNet. It was trained for 1,600 steps (8 epochs). Throughout the training process, the model demonstrated steady convergence and strong generalization capabilities. The training loss decreased significantly from an initial value of 0.7475 at step 50 to 0.0613 at step 1,600, while the validation loss dropped from 0.5478 to 0.251 over the same period. This consistent reduction in both losses highlights the model’s ability to minimize error effectively without overfitting. Furthermore, the accuracy improved from around 83% at the early stage to over 93% at the final step, reflecting robust performance on the validation set.

Error analysis

Table 7 presents examples of misclassified samples. Analysis revealed that most errors stemmed from subtle linguistic nuances and implicit intentions within the text, which made classification difficult. In addition, several samples contained writing issues such as typos, missing prepositions, or concatenated words without proper

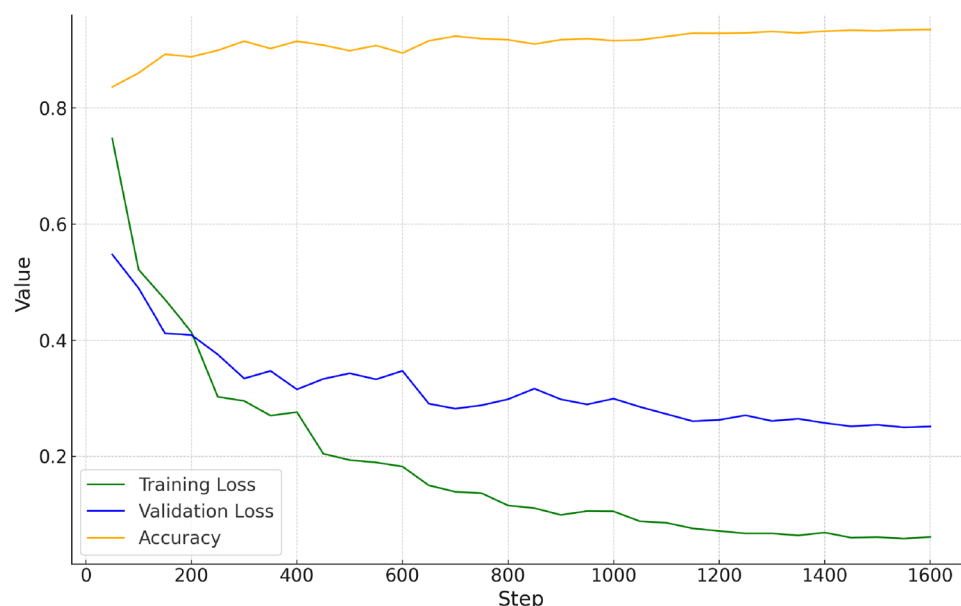


Fig. 6. Training metrics of the proposed hybrid model SaudiBERT-BiGRU-CapsNet.

Class	True label	Predicted	Original text	Translation
Sport	Pessimistic	Neutral	ياسر المسحل اخر واحد يفكر في كأس اسيا	Yasser Al-Misehal is the last person to think about the Asian Cup.
Sport	Pessimistic	Neutral	وين رايح يا ماني	Where are you going, Mané?
Sport	Pessimistic	Neutral	وفاز بصعوبة في اغلب المباريات مع الدفدة بالتحكيم	He won most of the matches with difficulty and some help from the referees.
Sport	Neutral	Optimistic	من المواضيع المحببة لي ايام الدراسة واثناء العمل هي العمل الجماعي	One of my favorite topics during my school days and work was teamwork.
Sport	Pessimistic	Neutral	ما عندنا مدرب هذا المختصر	We don't have a coach, that's the summary.
Finance	Pessimistic	Neutral	يبيع كميات طيب من اللي يشتريها لاحول ولا قوة الا بالله العلي العظيم	He's selling large quantities, but who's buying them? There is no power or trength except with Allah, the Most High, the Almighty.
Finance	Optimistic	Neutral	رابع شركة في التقييم من حيث المالية موصي عليها الراجحي	The fourth-ranked company in terms of financial evaluation, recommended by Al Rajhi.
Finance	Optimistic	Neutral	شركة البواني = بن لادن جديدة	Al-Bawani Company = The new Bin Laden.
Finance	Neutral	Optimistic	بعض المحللين يقول السقف العلوي للقناة الصاعدة 12200 والله اعلم	Some analysts say the upper ceiling of the rising channel is 12,200, and Allah knows best.
Finance	Neutral	Optimistic	هدفه 386 وهدف موجي 407	Its target is 386, and the wave target is 407.
Sport	Neutral	Pessimistic	اللاعبين بعيدين عن بعض	The players are far apart from each other.

Table 7. Misclassified samples: The translations capture both the literal meaning and the original intent of the Arabic text.

spacing. These factors introduced ambiguity and contextual challenges, particularly common in informal, dialect-rich social media text, and contributed to the model's misclassifications. Further analysis shows that many errors are related to implicit sentiment and domain-specific language. In the sports domain, idiomatic expressions and references to poor performance often lacked explicit negative tokens, causing the model to misclassify. Similarly, in the finance domain, factual descriptions of weak growth or unmet expectations were mislabeled. Furthermore, contrastive structures and negation often confounded the model. By prioritizing positive tokens over negative markers or contrastive cues, the classifier frequently defaulted to a neutral label when sentiment was not explicitly polarized. These findings indicate that the model struggles with implicit meaning, negation handling, and domain-specific expressions, which together explain the observed misclassifications.

Conclusion

Optimism and pessimism are critical emotional dimensions that often receive less attention in Arabic SA research. Optimism reflects an expectation of favorable outcomes and a positive outlook, whereas pessimism denotes a tendency to anticipate adverse outcomes and a negative perspective. Despite significant progress in general sentiment classification, there remains a substantial gap in effectively identifying and distinguishing these specific emotional states, particularly in the context of the Arabic language. To address this gap, we introduced a dedicated Arabic optimism and pessimism dataset, evaluated the performance of seven state-of-the-art Arabic language models, and proposed a novel hybrid model based on the BERT-BiGRU-CapsNet architecture. The results showed that SaudiBERT achieved the highest F1-Score of 93.11%, followed by MARBERTv2 with 87.25%. Moreover, the proposed hybrid model, SaudiBERT-BiGRU-CapsNet, outperformed all baseline models. By combining deep semantic feature extraction from BERT, sequential dependency modeling through BiGRU, and hierarchical relationship preservation via CapsNet, the hybrid approach enabled more effective sentiment classification, resulting in improved accuracy for distinguishing optimism and pessimism in Arabic text. The outcomes of this research provide important insights into the effectiveness of current Arabic NLP models for nuanced sentiment classification. By establishing a novel dataset and proposing a hybrid architecture, this work lays a solid foundation for future studies in context-aware language understanding and SA within the Arabic domain. Accurate identification of optimism and pessimism carries significant implications for enhancing individual well-being, predicting behavioral trends, and supporting the design of targeted mental health and social interventions. Beyond these applications, the findings also might offer value to businesses by enabling the automatic detection of optimism and pessimism in customer posts. Such fine-grained sentiment classification can guide marketing managers in analyzing customer behavior, supporting strategic decision-making, product development, and customer engagement.

This study has several limitations, most notably the reliance on text collected exclusively from the Hawamr and Kooora forums. Such a narrow data source may not fully capture the linguistic and stylistic diversity present across other Arabic online platforms. As both forums represent distinct community cultures with specialized interaction norms, primarily centered on finance and sports, the resulting dataset may reflect patterns that are not representative of broader Arabic discourse. Therefore, the findings should be interpreted within the specific context of these platforms, and no claims of generalizability are made across other domains, genres, or the wide range of Arabic dialectal variations. Another limitation of this study is the reliance on single-annotator labeling. While multi-annotator validation is widely regarded as best practice in benchmark dataset construction, the annotation in this work was conducted by a single domain-informed annotator following a detailed and predefined protocol. This decision was primarily driven by budgetary constraints, which play a significant role in annotation design. High-quality linguistic annotation, particularly for dialectal data, requires trained annotators and extensive quality control processes. Implementing multi-annotator schemes (e.g., double annotation with adjudication) substantially increases costs due to duplicated effort and the involvement of expert reviewers. In addition, time constraints further necessitated the adoption of a single-annotation strategy. Multi-annotator frameworks require coordination, iterative agreement assessments, and adjudication cycles, all of which extend the annotation timeline. To mitigate this, the annotation guidelines were iteratively refined through pilot annotation rounds, and consistency checks were conducted across temporally separated batches to reduce subjective drift. Moreover, the annotator is a native Arabic speaker from Saudi Arabia with high proficiency in the Saudi dialect. This linguistic and cultural background enabled the reliable capture of nuanced expressions of optimism and pessimism in the text. Also, a random review process was conducted to ensure the accuracy and correctness of the classifications. Additionally, the annotator has extensive experience in dataset annotation. From a methodological perspective, this study is also limited by the absence of fine-grained statistical validation and component-level analysis. Although the authors assess overall statistical significance across models using ANOVA, the work lacks pairwise statistical testing (e.g., paired t-tests or McNemar's tests) between the proposed hybrid model and baseline models. As a result, the statistical significance of the observed performance difference between SaudiBERT and the SaudiBERT-BiGRU-CapsNet model cannot be conclusively established. Furthermore, the absence of an ablation study to isolate the individual contributions of the BiGRU and CapsNet components limits the ability to precisely attribute performance gains to specific architectural elements. Therefore, we frame this component as an exploratory architectural extension rather than a definitively superior alternative.

To address these limitations, future research should expand the dataset to include additional domains, multiple online platforms, and a broader spectrum of Arabic dialects in order to enhance sociolinguistic diversity and generalizability. Increasing the number of pessimistic samples will also be necessary to mitigate class imbalance and improve model robustness. From a methodological standpoint, future work will incorporate pairwise statistical testing to rigorously evaluate model-level differences and conduct comprehensive ablation studies (e.g., SaudiBERT+BiGRU, SaudiBERT+CapsNet, and the full hybrid model) to better understand the contribution of each component. Despite these limitations, this research provides practical guidance for advancing context-aware sentiment analysis in Arabic NLP and outlines promising directions for future investigation, particularly in expanding domain coverage and addressing cultural and dialogic variation within the Arabic language.

Data availability

The datasets generated during and/or analysed during the current study are available in the Github repository, <https://github.com/FaisalQarah/Saudi-Optimism-Pessimism-Detection>

Code availability

The code files, along with the results, are available online on the zenodo repository, <https://zenodo.org/records>

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Author contributions

Conceptualization, F.Q. and T.A.; Methodology, F.Q.; Software, F.Q.; Validation, F.Q. and T.A.; Formal analysis, F.Q. and T.A.; Investigation, T.A.; Data curation, F.Q.; Writing—original draft, T.A. and F.Q.; Writing—review and editing, T.A. and F.Q. All authors have read and agreed to the published version of the manuscript.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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